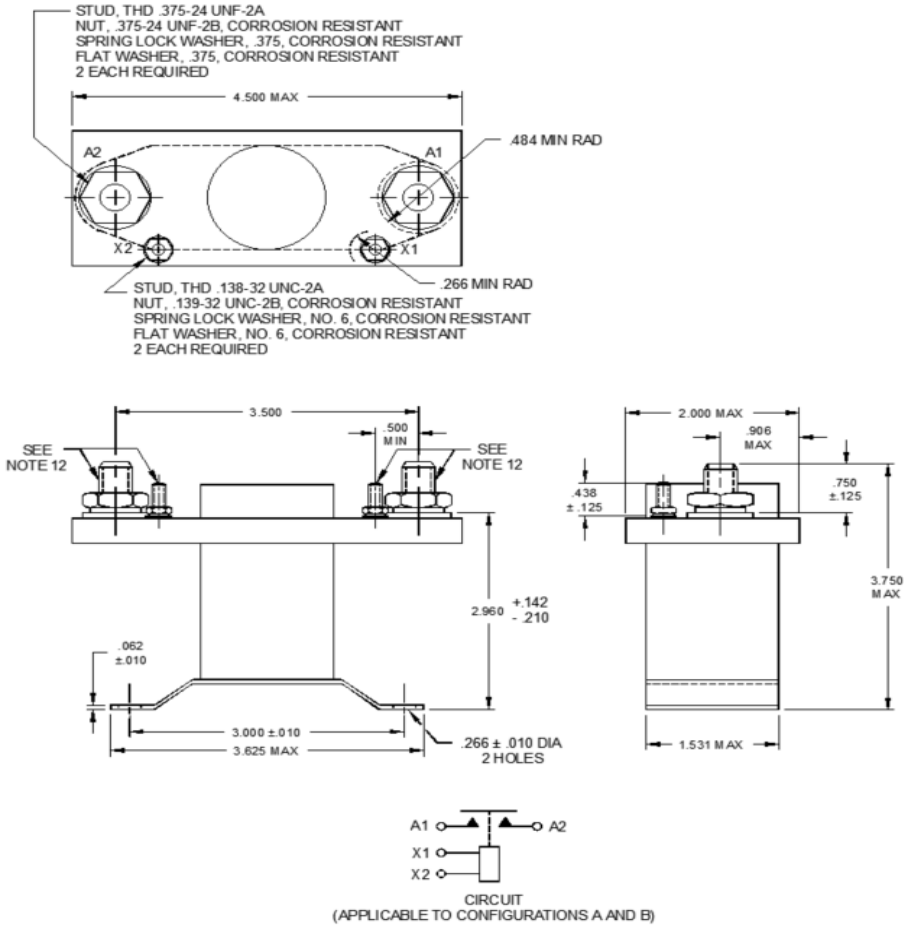
INCH-POUND
MS24172K
w/AMENDMENT 1
18 July 2025
SUPERSEDING
MS24172K
11 June 2020

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 200 AMPERES, 1 PST (N.O.), TYPE III, NONHERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

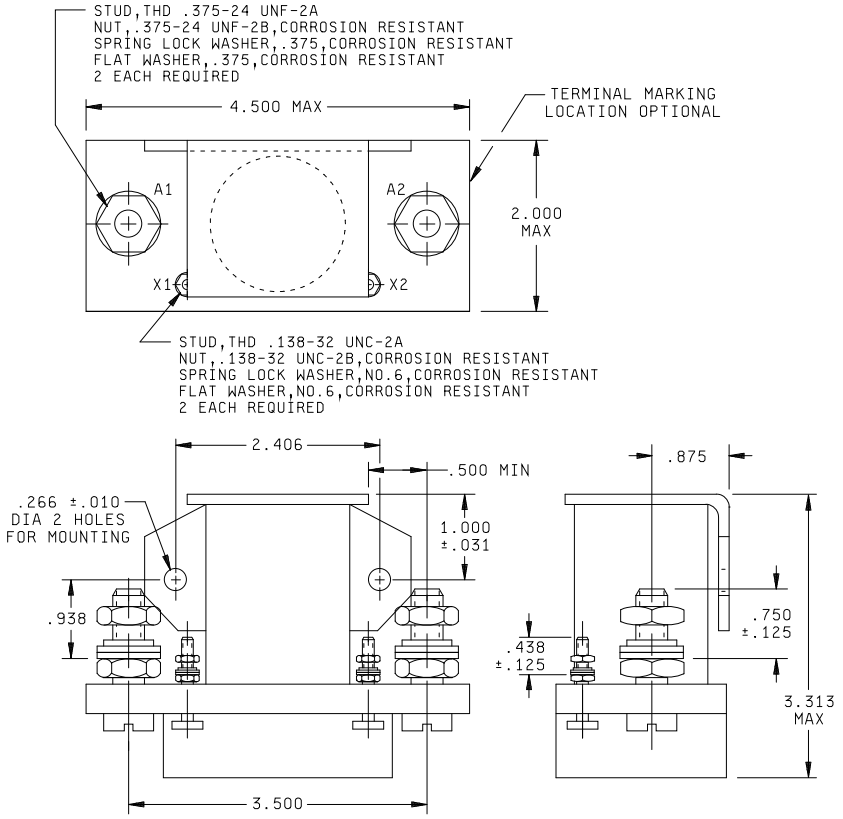


CONFIGURATION A

FIGURE I. Outline drawing (for details see tables I and II).

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CONFIGURATION B

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.010	0.25	.375	9.53	.906	23.01	3.000	76.20
.062	1.57	.438	11.13	.938	23.83	3.313	84.15
.125	3.18	.484	12.29	1.000	25.40	3.500	88.90
.138	3.51	.500	12.70	1.531	38.89	3.750	95.25
.142	3.61	.750	19.05	2.000	50.80	4.500	114.30
.210	5.33	.813	20.65	2.406	61.11		
.266	6.76	.875	22.23	2.960	75.18		

NOTES:
1. Dimensions are in inches.
2. Terminal cover not required.
3. Additional flat washer may be used for terminal seat.
4. Part number MS24172-D1 replaces part no. MS24172-1.
5. This specification sheet takes precedence over documents referenced herein.
6. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this specification sheet to the extent specified herein.
7. Metric equivalents are given for general information only.
8. Unless otherwise specified, tolerances are ±.062 (1.575 mm).
9. Terminal temperature rise under continuous current conditions, 95°C. Intermediate current to be conducted at 71°C.
10. Shape of relay optional within envelope dimensions.
11. Cadmium or cadmium compounds are prohibited on external hardware.
12. Bolt heads for terminals may extend below their mounting surface.

FIGURE I. Outline drawing (for details see tables I and II) - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number MS24172-	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1 1/	III	dc	Stud	Bracket - bottom	1.33
D2 2/	III	dc	Stud	Bracket - side	1.33

1/ For Government logistics support MS24172-D1 shall be used in lieu of AN3371-2.
2/ For Government logistics support MS24172-D2 shall be used in lieu of AN3371-1.

TABLE II. Operating characteristics.

PIN MS 24172-	Coil data											Time - (milliseconds maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Bounce			
														Main		Aux	
		Volts 1/	Freq Hz	Res Ω +15% -10	Volts	Amp	Normal 2/	High temp. test	Cont current test					NO	NC	NO	NC
D1	X1,X2	28	dc	10	29	4.5	7.5	11.5	---	3.0	.5	25	18	5.0	---	---	---
D2	X1,X2	28	dc	10	29	4.5	7.5	11.5	---	3.0	.5	25	18	5.0	---	---	---

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	50	200											
Inductive	10	100											
Motor	50	200											
Lamp													
Transfer load													2/
Mechanical life reduced current	100	50											
Mixed loads													

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Environmental characteristics:

Temperature range	-55°C to +71°C
Maximum altitude rating	50,000 ft
Shock G-level	25 g's
Duration	6-9 ms
Max duration contact opening	2 ms
Vibration - sinusoidal (see table IV)	
Vibration - Random	N/A
High shock	N/A
Acceleration	10 g's

TABLE IV. Vibration levels.

5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1,500 Hz
.08 DA	.06 DA	2 g's	2 g's	

Electrical characteristics:

Insulation resistance, initial:	100 megohms
After life or environmental tests:	50 megohms
Dielectric strength (sea level):	2-5 seconds

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250 V rms	N/A	1,000 V rms	N/A
Aux contacts	1,250 V rms	N/A	1,000 V rms	N/A
All other points	1,250 V rms	N/A	1,000 V rms	N/A

Dielectric strength (altitude): 1 minute

	28 V dc	115 V ac
Coil to case	500 V rms	N/A
Aux contacts	500 V rms	N/A
All other points	700 V rms	N/A

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Max contact drop initial:	0.150 volt
After life test:	0.175 volt
Overload current	(NO) 1,600 amperes
Rupture current	(NO) 2,000 amperes
Duty rating	Intermittent
RFI spec	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Qualification by similarity: See MIL-PRF-6106.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-461

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing Activity
DLA - CC

(Project 5945-2025-030)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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